

Edexcel International AS Chemistry



Your notes

Group 7

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- * Halogen Displacement Reactions
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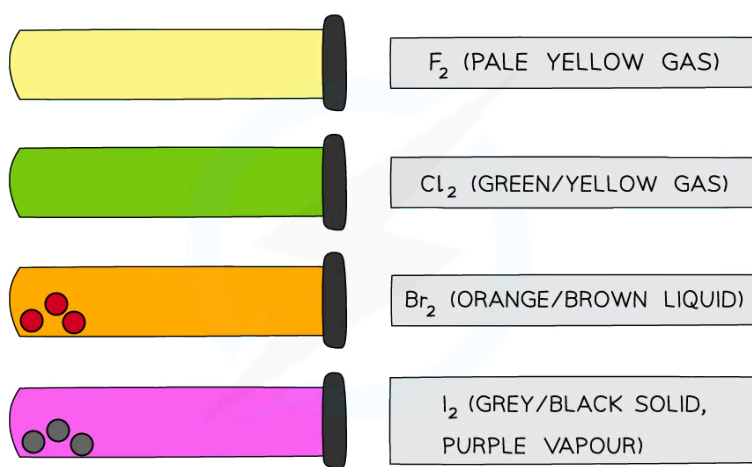


Trends of the Group 7 Elements

- The Group 7 elements are called **halogens**
- The halogens have uses in water purification and as bleaching agents (chlorine), as flame-retardants and fire extinguishers (bromine) and as antiseptic and disinfectant agents (iodine)

Colours

- All halogens have distinct **colours** which get **darker** going down the group



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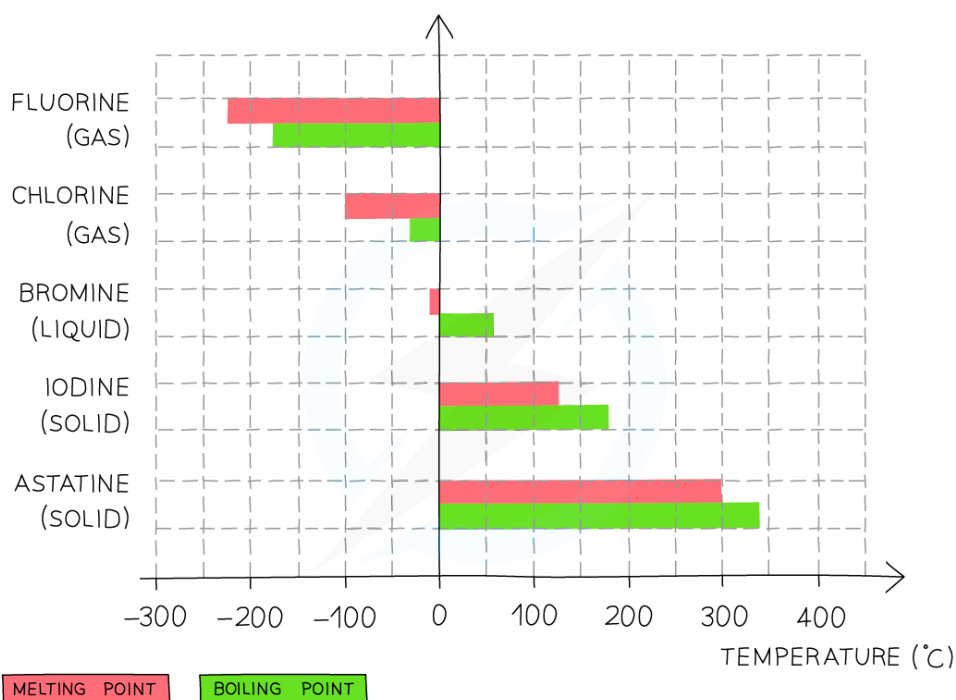
The colours of the Group 7 elements get darker going down the group

Volatility

- **Volatility** refers to how easily a substance can evaporate
 - A volatile substance will have a low boiling point



Your notes



The melting and boiling points of the Group 7 elements increase going down the group which indicates that the elements become less volatile

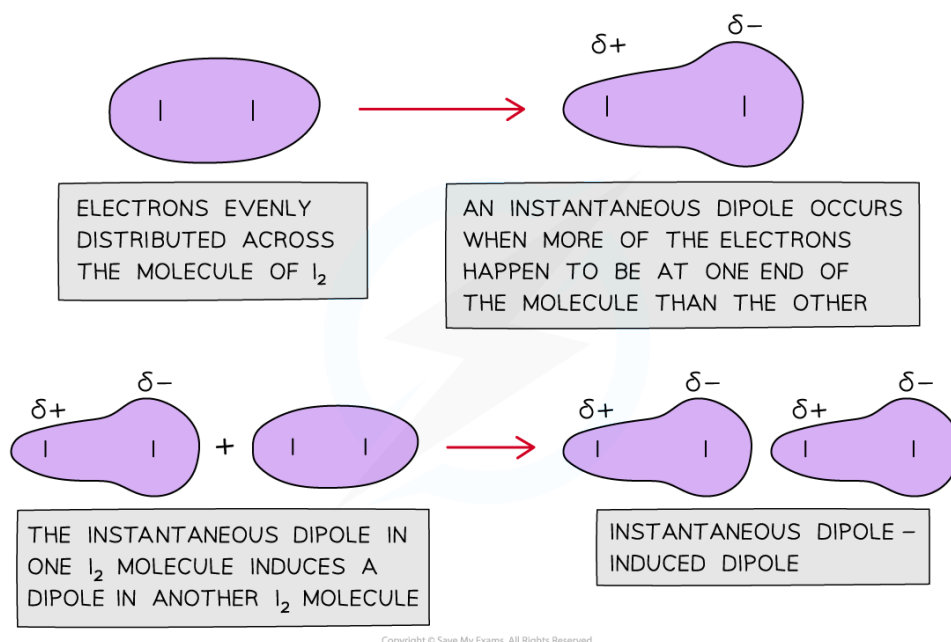
- Going down the group, the **boiling point** of the elements increases which means that the **volatility** of the halogens decreases
 - This means that fluorine is the most volatile and iodine the least volatile

Trend in melting and boiling points

- Halogens are non-metals and are **diatomic molecules** at room temperature
 - This means that they exist as molecules which are made up of two similar atoms, such as F_2
- The halogens are **simple molecular structures** with **weak** London dispersion forces between the diatomic molecules caused by instantaneous dipole-induced dipole forces



Your notes



The diagram shows that a sudden imbalance of electrons in a nonpolar molecule can cause an instantaneous dipole. When this molecule gets close to another non-polar molecule it can induce a dipole as the cloud of electrons repel the electrons in the neighbouring molecule to the other side

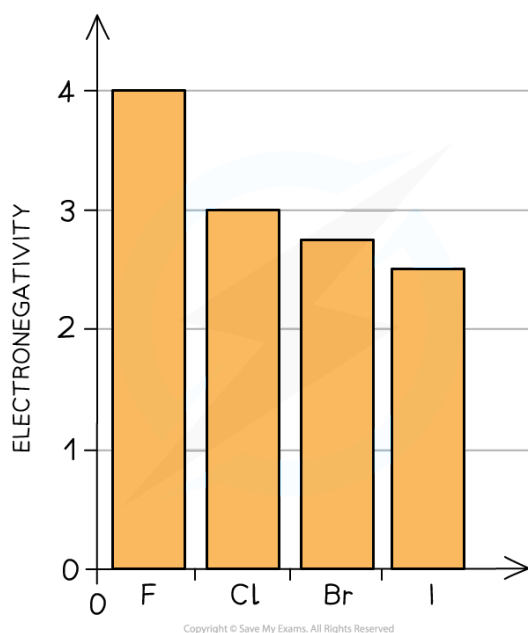
- The more **electrons** there are in a molecule, the greater the instantaneous dipole-induced dipole forces
- Therefore, the **larger** the molecule the **stronger** the London dispersion forces between molecules
- This is why as you go down the group, it gets more difficult to separate the molecules and the **melting** and **boiling points** increase
- As it gets more difficult to separate the molecules, the **volatility** of the halogens **decreases** going down the group

Trend in electronegativity

- The electronegativity of the halogens decreases down the group



Your notes

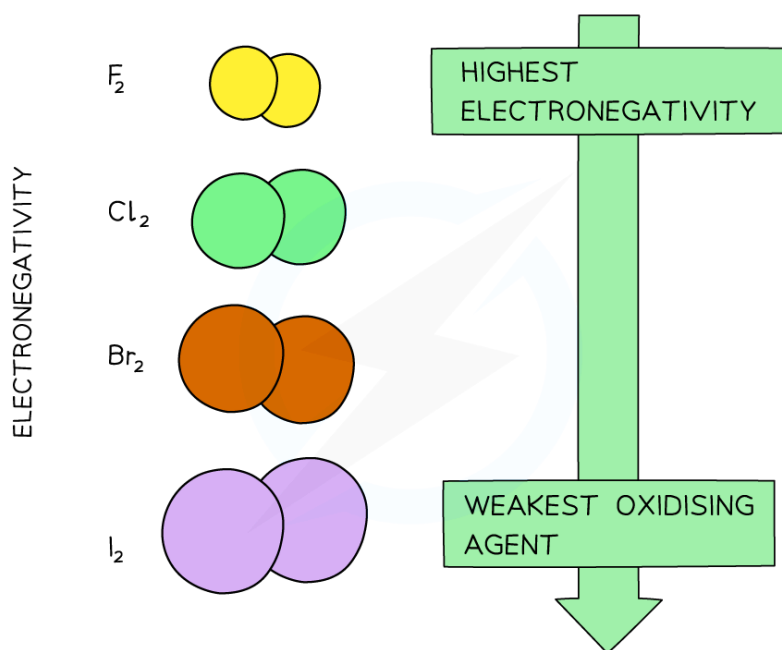


The electronegativity of the halogens decreases going down the group

- The **electronegativity** of an atom refers to how strongly it attracts electrons towards itself in a covalent bond
- The decrease in electronegativity is linked to the size of the halogens
- Going down the group, the atomic radii of the elements increase which means that the outer shells get further away from the nucleus
- An 'incoming' electron will therefore experience more **shielding** from the attraction of the positive nuclear charge
- The halogens' ability to accept an electron (their **oxidising power**) therefore decreases going down the group



Your notes



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With increasing atomic size of the halogens (going down the group) their electronegativity, and therefore oxidising power, decreases

Reactivity

- When a halogen atom reacts it will usually gain an electron, to form a $1-$ ion ($X + e^- \rightarrow X^-$)
- The oxidation number has decreased from 0 to -1 , therefore reduction has occurred
- Therefore halogens will act as oxidising agents
- Down Group 7 we have seen that the atoms become larger so the outer electrons are further away and are therefore more shielded from the positive nucleus
- Larger halogen atoms such as iodine will find it more difficult to attract incoming electrons needed to form the $1-$ ion
- Therefore the reactivity decreases down Group 7

Reaction with hydrogen

- To demonstrate the decrease in the reactivity we can look at the reaction with hydrogen gas
- The table outlines the trend in the reactivity of the halogens with hydrogen gas
- As we can see the reaction becomes less vigorous down the group

Reaction between Halogen & Hydrogen Gas



Your notes

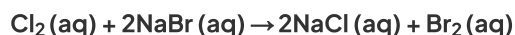
Equation for Reaction	Description of Reaction
$\text{H}_2(\text{g}) + \text{F}_2(\text{g}) \longrightarrow 2\text{HF}(\text{g})$	Reacts explosively even in cool, dark conditions
$\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \longrightarrow 2\text{HCl}(\text{g})$	Reacts explosively in sunlight
$\text{H}_2(\text{g}) + \text{Br}_2(\text{g}) \longrightarrow 2\text{HBr}(\text{g})$	Reacts slowly on heating
$\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$	Forms an equilibrium mixture on heating

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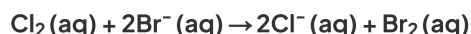


Halogen Displacement

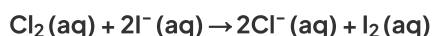
- The reactivity of halogens is also shown by their **displacement reactions** with other halide ions in solutions
 - A **more reactive** halogen can displace a **less reactive** halogen from a halide solution of the less reactive halogen
 - Eg. The addition of chlorine water to a solution of bromine water:



- The chlorine has displaced the bromine from solution as it is more reactive which can be summarised in the following ionic equation by removing the sodium spectator ions:



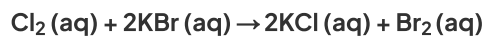
- Chlorine can also displace iodine from a solution of iodide ions:



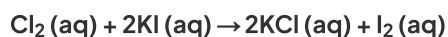
- We can see these are both redox reactions by taking a look at changes in the oxidation number of each element in the reaction
 - Br and I both change from $-1 \rightarrow 0$ so the bromine and iodine have been oxidised in their respective reactions
 - Cl $= 0 \rightarrow -1$ so the chlorine has been reduced in both reactions
 - No change in oxidation number for the sodium

Chlorine with Bromides & Iodides

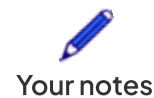
- If you add chlorine solution to colourless potassium bromide or potassium iodide solution a displacement reaction occurs:
 - The solution becomes **yellow-orange** as bromine is formed
 - The solution becomes **brown** as iodine is formed
- If an **organic solvent** is added, such as cyclohexane, the following observations are seen:
 - The organic layer will appear **yellow-orange** as bromine is formed
 - The organic layer will appear **purple** as iodine is formed
 - The organic solvent is useful as the halogens are more soluble in this layer which helps observe the colour changes more easily
- Chlorine is **above** bromine and iodine in Group 7 so it is more reactive
- Chlorine will **displace** bromine or iodine from an aqueous solution of the metal halide:



chlorine + potassium bromide $\rightarrow$ potassium chloride + bromine



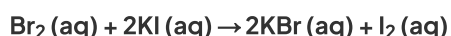
chlorine + potassium iodide $\rightarrow$ potassium chloride + iodine



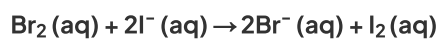
Bromine with Iodides

- Bromine is above iodine in Group 7 so it is **more** reactive
- Bromine will displace iodine from an aqueous solution of the metal iodide

bromine + potassium iodide $\rightarrow$ potassium bromide + iodine



- The reaction will turn **brown** as iodine is formed
- If an organic solvent is added the organic layer will appear **purple** as iodine is formed
- We can show that this is a redox reaction by looking at the changes in oxidation numbers:
 - $\text{I} = -1 \rightarrow 0$ so the iodine has been oxidised
 - $\text{Br} = 0 \rightarrow -1$ so the bromine has been reduced
 - No change in oxidation number for the potassium
- Rather than writing the full equation, we can also write the ionic equation by removing the potassium spectator ion





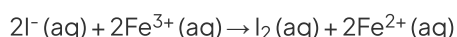
Redox Reactions of the Halogens

Reactions with Group 1 & 2 metals

- The halogens react with some metals to form **ionic compounds** which are **metal halide salts**
- In all reactions where halogens are reacting with metals, the metals are being oxidised
- Reaction of sodium and chlorine
 - $2\text{Na (s)} + \text{Cl}_2 \text{ (g)} \rightarrow 2\text{NaCl (s)}$
 - Na is being oxidised, the oxidation number is changing from 0 to +1
- Calcium is a group 2 metal:
 - $\text{Ca (s)} + \text{Br}_2 \text{ (l)} \rightarrow \text{CaBr}_2 \text{ (s)}$
 - Ca is being oxidised, the oxidation number is changing from 0 to +2
- Therefore the halogens are acting as oxidising agents

Reactions with Iron(II)

- Chlorine and bromine can oxidise iron(II) to iron(III)
$$\text{Cl}_2 \text{ (g)} + 2\text{Fe}^{2+} \text{ (aq)} \rightarrow 2\text{Cl}^- \text{ (aq)} + 2\text{Fe}^{3+} \text{ (aq)}$$
$$\text{Br}_2 \text{ (g)} + 2\text{Fe}^{2+} \text{ (aq)} \rightarrow 2\text{Br}^- \text{ (aq)} + 2\text{Fe}^{3+} \text{ (aq)}$$
- However, iodine is not a strong enough oxidising agent to oxidise iron(II) to iron(III)
- Iodine is actually oxidised from iodide ions to iodine by iron(III)



Disproportionation reaction

- A **disproportionation reaction** is a reaction in which the same species is both oxidised and reduced
- The reaction of **chlorine** with **dilute alkali** is an example of a disproportionation reaction
- In these reactions, the chlorine gets oxidised and reduced at the same time
- Different reactions take place at different **temperatures** of the dilute alkali

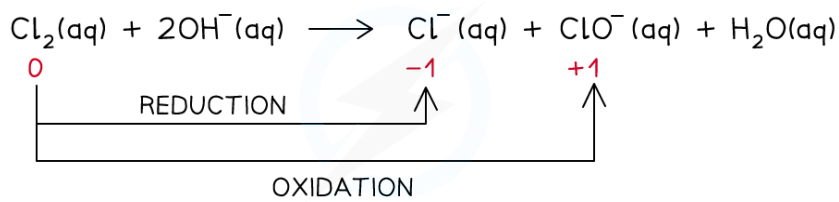
Chlorine in cold alkali (15 °C)

- The reaction that takes place is:



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- The ionic equation is:

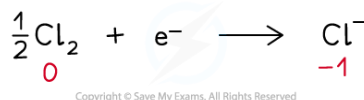


Your notes

- The ionic equation shows that the chlorine gets both oxidised and reduced
- Chlorine gets oxidised as there is an increase in ox. no. from 0 to +1 in $\text{ClO}^-(\text{aq})$
 - The half-equation for the oxidation reaction is:



- Chlorine gets reduced as there is a decrease in ox. no. from 0 to -1 in $\text{Cl}^-(\text{aq})$
 - The half-equation for the reduction reaction is:

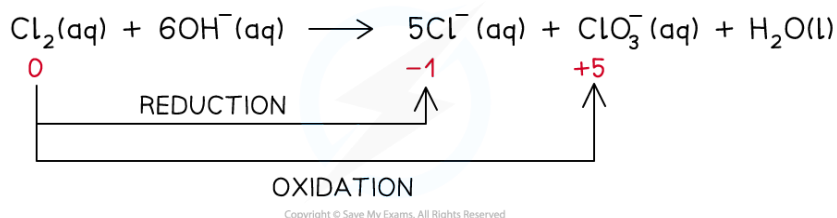


Chlorine in hot alkali (70 °C)

- The reaction that takes place is:



- The ionic equation is:



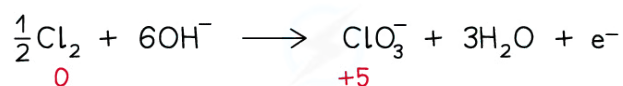
- The ionic equation shows that the chlorine gets both oxidised and reduced



Your notes

- Chlorine gets oxidised as there is an increase in ox. no. from 0 to +5 in $\text{ClO}_3^-(\text{aq})$

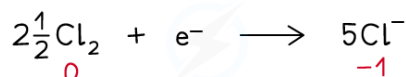
- The half-equation for the oxidation reaction is:



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- Chlorine gets reduced as there is a decrease in ox. no. from 0 to -1 in $\text{Cl}^-(\text{aq})$

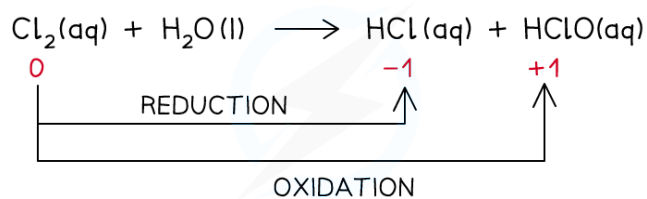
- The half-equation for the reduction reaction is:



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Drinking water

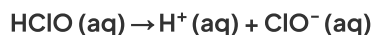
- Chlorine can be used to clean water and make it drinkable
- The reaction of chlorine in water is a **disproportionation reaction** in which the chlorine gets both **oxidised** and **reduced**



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The disproportionation reaction of chlorine with water in which chlorine gets reduced to HCl and oxidised to HClO

- Chloric(I) acid (HClO) **sterilises** water by killing **bacteria**
- Chloric acid can further dissociate in water to form $\text{ClO}^-(\text{aq})$:



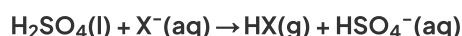
- $\text{ClO}^-(\text{aq})$ also acts as a **sterilising agent** cleaning the water



Other Reactions of the Halogens

Concentrated sulfuric acid

- Chloride, bromide and iodide ions react with concentrated sulfuric acid to produce **toxic gases**
- These reactions should therefore be carried out in a fume cupboard
- The general reaction of the halide ions with concentrated sulfuric acid is:

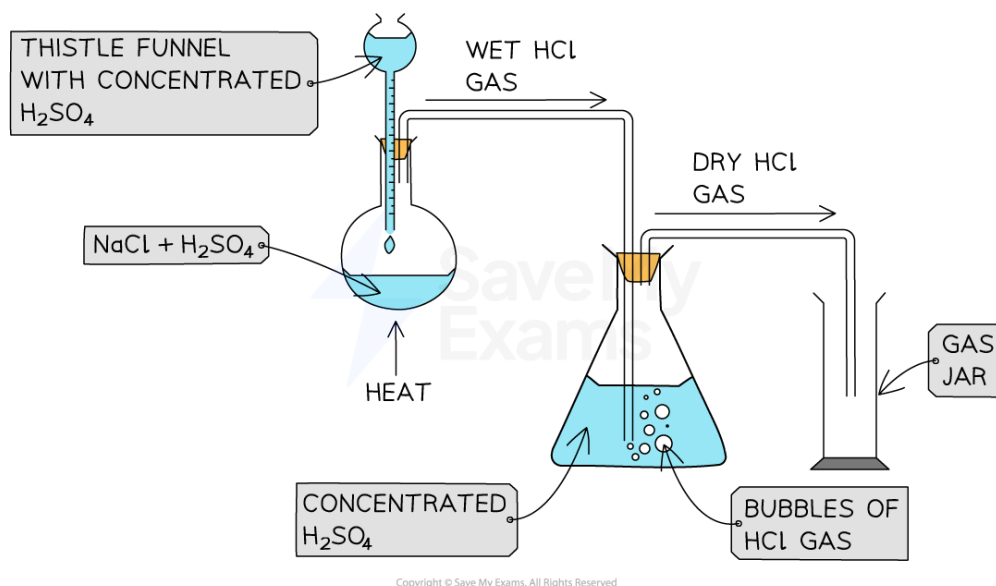


(general equation)

Where X^- is the halide ion

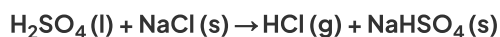
Reaction of chloride ions with concentrated sulfuric acid

- Concentrated sulfuric acid is dropwise added to sodium chloride crystals to produce **hydrogen chloride gas**



Apparatus set up for the reaction of sodium chloride with concentrated sulfuric acid

- The reaction that takes place is:

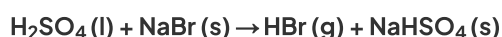


- The HCl gas produces is seen as **misty fumes**

Reaction of bromide ions with concentrated sulfuric acid



- The **thermal stability** of the hydrogen halides decreases down the group
- The reaction of sodium bromide and concentrated sulfuric acid is:



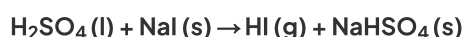
- The concentrated sulfuric acid **oxidises** HBr which decomposes into **bromine** and **hydrogen gas** and sulfuric acid itself is **reduced** to **sulfur dioxide gas**:



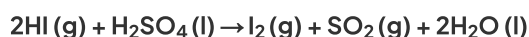
- The bromine is seen as a **reddish-brown gas**

Reaction of iodide ions with concentrated sulfuric acid

- The reaction of sodium iodide and concentrated sulfuric acid is:



- **Hydrogen iodide** decomposes the **easiest**
- Sulfuric acid oxidises the hydrogen iodide to several extents:
- The concentrated sulfuric acid **oxidises** HI and is itself **reduced** to **sulfur dioxide gas**:



- Iodine is seen as a violet/purple vapour
- The concentrated sulfuric acid **oxidises** HI and is itself **reduced** to **sulfur**:



- Sulfur is seen as a **yellow solid**
- The concentrated sulfuric acid **oxidises** HI and is itself **reduced** to **hydrogen sulfide**:



- Hydrogen sulfide has a **strong smell of bad eggs**

Further redox reaction involving sulfur dioxide

- HI is such a strong reducing agent that it can reduce sulfur dioxide, SO_2 , further to hydrogen sulfide, H_2S



- Sulfur is reduced from +4 in SO_2 to -2 in H_2S , a gain of 6 electrons:
 - This demonstrates reduction
- The iodide ion is oxidised to iodine (I_2), seen as purple vapour
- The bad egg smell of H_2S confirms the presence of this gas
- This final step highlights the maximum reducing power of iodide ions, not seen with Cl^- or Br^-

Halide ion Reactions with Concentrated Sulfuric Acid Table



Your notes

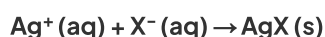
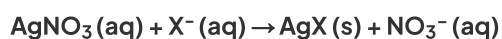
Halide ion	Reaction with conc H_2SO_4	Observations
$\text{Cl}^- (\text{aq})$	$\text{H}_2\text{SO}_4 (\text{l}) + \text{NaCl} (\text{s}) \rightarrow \text{HCl} (\text{g}) + \text{NaHSO}_4 (\text{s})$	Misty fumes of HCl gas
$\text{Br}^- (\text{aq})$	$\text{H}_2\text{SO}_4 (\text{l}) + \text{NaBr} (\text{s}) \rightarrow \text{HBr} (\text{g}) + \text{NaHSO}_4 (\text{s})$ $\text{H}_2\text{SO}_4 (\text{l}) + 2\text{HBr} (\text{g}) \rightarrow \text{Br}_2 (\text{g}) + \text{SO}_2 (\text{g}) + 2\text{H}_2\text{O} (\text{l})$	Misty fumes of HBr gas Chocking gas of SO_2 and reddish brown gas of Br_2
$\text{I}^- (\text{aq})$	$\text{H}_2\text{SO}_4 (\text{l}) + \text{NaI} (\text{s}) \rightarrow \text{HI} (\text{g}) + \text{NaHSO}_4 (\text{s})$ $\text{H}_2\text{SO}_4 (\text{l}) + 2\text{HI} (\text{g}) \rightarrow \text{I}_2 (\text{s}) + \text{SO}_2 (\text{g}) + 2\text{H}_2\text{O} (\text{l})$ $\text{H}_2\text{SO}_4 (\text{l}) + 6\text{HI} (\text{g}) \rightarrow 3\text{I}_2 (\text{s}) + \text{S} (\text{s}) + 4\text{H}_2\text{O} (\text{l})$ $\text{H}_2\text{SO}_4 (\text{l}) + 8\text{HI} (\text{g}) \rightarrow 4\text{I}_2 (\text{s}) + \text{H}_2\text{S} (\text{g}) + 4\text{H}_2\text{O} (\text{l})$	Misty fumes of HI gas Chocking gas of SO_2 and purple vapour of I_2 Yellow solid of $\text{S} (\text{s})$ Strong, bad egg smell of $\text{H}_2\text{S} (\text{g})$

Explaining the Trend in Reducing Ability of Halide Ions

- As you go down Group 7, the halide ions get bigger ($\text{Cl}^- \rightarrow \text{Br}^- \rightarrow \text{I}^-$)
- So, the attraction between the outer electrons and the nucleus gets weaker
- This makes it easier for the larger halide ions to lose electrons
- When they lose electrons, they reduce other substances,
- So the reducing power increases down the group:
 - $\text{Cl}^- < \text{Br}^- < \text{I}^-$

Silver ions & ammonia

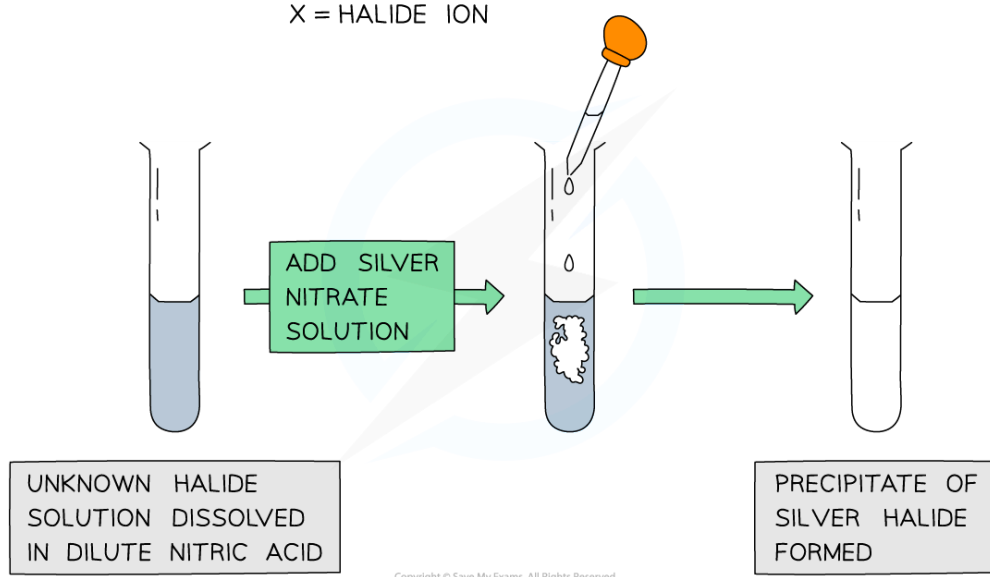
- Halide ions can be identified in an **unknown solution** by dissolving the solution in **nitric acid** and then adding a **silver nitrate solution** followed by **ammonia** solution
- The halide ions will react with the silver nitrate solution as follows:



- X^- is the halide ion in both equations
- If the unknown solution contains halide ions, then a **precipitate** of the **silver halide** will be formed (AgX)



Your notes



A silver halide precipitate is formed upon addition of silver nitrate solution to halide ion solution

- **Dilute** followed by **concentrated ammonia** is added to the silver halide solution to identify the halide ion
- If the precipitate dissolves in **dilute** ammonia the unknown halide is **chloride**
- If the precipitate does not dissolve in dilute but in **concentrated** ammonia the unknown halide is **bromide**
- If the precipitate does not dissolve in **dilute** nor **concentrated** ammonia the unknown halide is iodide



Your notes

AgCl PRECIPITATE

AgCl FORMS A SOLUBLE COMPLEX WITH DILUTE AMMONIA

AgBr PRECIPITATE

AgBr FORMS A SOLUBLE COMPLEX WITH CONCENTRATED AMMONIA

AgI PRECIPITATE

AgI DOESN'T FORM A SOLUBLE COMPLEX WITH AMMONIA

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Silver chloride and silver bromide precipitates dissolve on addition of ammonia solution whereas silver iodide doesn't

Reaction of Halide Ions with Silver Nitrate & Ammonia Solutions Table

Halide ion	Colour of Silver Halide Solution	Effect of adding dil. NH_3	Effect of adding conc. NH_3
$\text{Cl}^- (\text{aq})$	White	Dissolves	Dissolves
$\text{Br}^- (\text{aq})$	Cream	Insoluble	Dissolves
$\text{I}^- (\text{aq})$	Pale yellow	Insoluble	Insoluble

Reactions with hydrogen halides



Your notes

- When a halogen reacts with hydrogen a hydrogen halide is produced, for example:
 - $\text{Cl}_2(\text{g}) + \text{H}_2(\text{g}) \rightarrow 2\text{HCl}(\text{g})$
- These hydrogen halides react with ammonia gas to form **ammonium halides**
 - $\text{NH}_3(\text{g}) + \text{HCl}(\text{g}) \rightarrow \text{NH}_4\text{Cl}(\text{s})$
- Hydrogen halides will also react with water
- For example, hydrogen chloride also dissolves in water to form **hydrochloric acid**
 - $\text{HCl}(\text{g}) \rightarrow \text{H}^+(\text{aq}) + \text{Cl}^-(\text{aq})$

Fluorine & Astatine

Fluorides and Astatides

- If you are asked to predict reactions of fluorides and astatides that you are not familiar with you can use the information about Group 7 to help
- There are, however, exceptions to some of the reactions
 - For example, silver fluoride is soluble, so we can not use acidified silver nitrate to test for the presence of fluoride ions
 - $\text{Ag}^+(\text{aq}) + \text{F}^-(\text{aq}) \rightarrow \text{AgF}(\text{aq})$



Worked Example

Write an equation to show the formation of an acid when hydrogen astatide is added to water.

Answer:

- Based on the reaction of hydrogen iodide with water:
 - $\text{HI}(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightarrow \text{H}_3\text{O}^+(\text{aq}) + \text{I}^-(\text{aq})$
 - So, hydrogen astatide must be:
 - $\text{HAt}(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightarrow \text{H}_3\text{O}^+(\text{aq}) + \text{At}^-(\text{aq})$